INEQUALITIES AND LIMIT THEOREMS

MARKOV'S INEQUALITY X r.v., g non decreasing, non-neg.

used to prove

$$P(\{|X|>c\}) \leq \frac{E[g(|X|)]}{g(c)} \quad \forall c>0$$

CHEBYSHEV'S INEQUALITY X r.v. with $E[X]$, $Var(X)$ finite

equiv. results

$$P(\{|X-E[X]|>c\}) \leq \frac{Var(X)}{c^2} \quad \forall c>0$$

$$P(\{|X-E[X]|>k\cdot SD(X)\}) \leq \frac{1}{k^2} \quad \forall k>0$$

WLLN X_i i.i.d. $X_i \sim P_X$ and $E[X]$ finite

$$\frac{1}{n}\sum X_i \longrightarrow E[X_i]$$

CLT X_i i.i.d. $X_i \sim P_X$ $E[X_i]$, $Var(X_i)$ finite

ARBITRARY DISTRIBUTIONS (for big n)

1. $\frac{1}{n}\sum X_i \sim N\left(E[X_i], \frac{Var(X_i)}{n}\right)$ approximately
2. $\dfrac{\frac{1}{n}\sum X_i - E[X_i]}{SD(X_i)/\sqrt{n}} \sim N(0,1)$ approximately
3. $\sum X_i \sim N(nE[X_i], nVar(X_i))$ approximately

equivalent results